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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

AEROSPACE AVIONICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Explain VHF and UHF communications.	L2	CO1	**02**
	b.	Explain the duty cycle.	L2	CO2	**02**
	c.	Differentiate between great circle and rhumb line navigation.	L2	CO3	**02**
	d.	Explain ADS-B and EGPWS in aircraft navigation systems.	L2	CO5	**02**
	e.	Differentiate between atomic clocks and signal correction in GPS.	L2	CO6	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain the challenges in aircraft communication.	L2	CO1	**05**
		OR			
	b.	Illustrate the applications of fiber optic communication.	L2	CO1	**05**
3.	**a.**	Discuss the advantages and limitations of primary surveillance RADAR.	L2	CO2	**05**
		OR			
	b.	Explain the applications of secondary surveillance RADAR	L2	CO2	**05**
4.	**a.**	Explain the MIL-STD-15538 and ARINC 429 avionics buses..	L2	CO3	**05**
		OR			
	b.	Explain dead reckoning.	L2	CO4	**05**
5.	**a.**	Illustrate the followings: **i.** Flight control systems. **ii.** Navigation systems. **iii.** Communication systems. **iv.** Surveillance systems. **v.** Flight instruments.	L2	CO5	**05**
		OR			
	b.	Illustrate GNSS and INS in aircraft navigation systems.	L2	CO5	**05**
6.	**a.**	Explain GPS in air, surface and space navigation.	L2	CO6	**05**
		OR			
	b.	Explain the working of GPS.	L2	CO6	**05**
7.	**a.**	Discuss the advantages and application of ACARS.	L2	CO1	**05**
		OR			
	b.	Explain peak transmitter power and average transmitter power.	L2	CO2	**05**

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		PART-C			6 X 10=60
8.	a.	Discuss the key features, advantages and disadvantages of digital communication	L2	CO1	10
		OR			
	b.	Discuss the advantages, disadvantages and applications of selective calling.	L2	CO1	10
9.	a.	Examine the maximum unambiguous range.	L3	CO2	10
		OR			
	b.	Explain the applications of RADAR.	L3	CO2	10
10.	a.	Describe the relation between Maps and Geodesy.	L3	CO3	10
		OR			
	b.	Discuss the key features and the applications of WGS-85 system.	L3	CO3	10
11.	a.	Describe distance measuring equipment and instrument landing system.	L3	CO4	10
		OR			
	b.	Explain automatic direction finder and VHF omnidirectional range.	L3	CO4	10
12.	a.	Explain the avionics buses.	L3	CO5	10
		OR			
	b.	Illustrate the functional blocks of AFDX with a neat block diagram.	L3	CO5	10
13.	a.	Illustrate INS-GPS system with a neat block diagram.	L3	CO6	10
		OR			
	b.	Examine the key functions, advantages, challenges and applications of GPS in space.	L3	CO6	10